

KE XING



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EDUCATION

Ph.D. Student, Beijing Jiaotong University, China

Computer Science and Technology

Research interests in 3D Vision and Video World Model.

2026.09 – Expected 2031

Advisor: Prof. Xiaojie Jin

B.S. Student, Beijing Jiaotong University, China

Computer Science and Technology

2022.09 – 2026.06

Advisor: Prof. Xiaojie Jin

RESEARCH EXPERIENCE

Stereoscopic Video Generation

2025.08 – 2025.11

High demand for quality stereo video in XR clashes with the high cost and artifacts of existing methods. We present StereoWorld, an end-to-end framework that turns monocular videos into high-fidelity stereo pairs. It uses a geometry-aware pretrained video generator for accurate 3D structure and spatio-temporal tiling for efficient high-resolution synthesis. We also construct a large stereo video dataset with over 11 million IPD-aligned frames.

4D Panoramic Scene Generation and Reconstruction

2024.02 – 2025.07

As VR/AR advances rapidly, demand for high-quality immersive dynamic scenes has grown. We propose a dual-branch panoramic video generation framework: a perspective branch for local detail and diversity, and a panorama branch for global consistency. We further extend it to 4D via spatio-temporal alignment reconstruction for coherent spatial-temporal representation.

Protein Sequence Classification

2023.04 – 2024.04

Inspired by the remarkable sequence processing capabilities of pre-trained NLP models like BERT, we investigated applying BERT, pre-trained on large-scale protein sequence datasets and fine-tuned on domain-specific data, to address protein sequence classification within specific domains.

Video Distortion Correction

2024.05 – 2025.04

Developed a diffusion-based framework for wide-angle video distortion correction by learning image-level optical flow and transferring it to video via pseudo-labeling and temporal consistency, effectively addressing limited labeled data.

PUBLICATIONS

- **Ke Xing**, Longfei Li, Yuyang Yin, Hanwen Liang et al. “StereoWorld: Geometry-Aware Monocular-to-Stereo Video Generation” **CVPR, 2026, CCF-A**.
- **Ke Xing***, Hanwen Liang*, Dejie Xu, Yuyang Yin, Konstantinos N. Plataniotis, Yao Zhao, Yunchao Wei. “TiP4GEN: Text to Immersive Panorama 4D Scene Generation.” **ACM Multimedia, 2025, CCF-A**.
- Wenbo Nie, Lang Nie, Chunyu Lin, Jingwen Chen, **Ke Xing**, Jiyuan Wang et al. “Beyond Wide-Angle Images: Structure-to-Detail Video Portrait Correction via Unsupervised Spatiotemporal Adaptation.” **AAAI, 2026, CCF-A**.
- Ana Martínez Gascueña, Haiyang Wu, Rui Wang, C David Owen, Pedro J Hernando, Serena Monaco, Matthew Penner, **Ke Xing** et al. “Exploring the sequence-function space of microbial fucosidases.” **Nature Communications Chemistry, 2024**.

INTERNSHIP EXPERIENCE

Dzine.ai, China

2025.08 – 2026.04

Research Topics: Stereoscopic Video Generation for XR/VR Devices